

Indoor High Precision Positioning System Based on Visible Light Communication and Location Fingerprinting

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Abstract—Indoor positioning system based on visible light communication and location fingerprinting can achieve cm-level positioning accuracy, which has become an important candidate for high precision positioning. However, in the case of non-uniform and low-density location fingerprinting distribution, it will still lead to large errors and poor robustness. To solve this problem, an accurate visible light positioning (VLP) method based on location fingerprinting and meta-heuristic is proposed, and all possible cases of uneven distribution of location fingerprinting are considered by introducing the random selection rate. The effectiveness of the proposed VLP method is compared with other six VLP methods based on location fingerprinting. Simulation results show that when the signal-to-noise ratio is 20 dB, the average positioning error of the proposed VLP method is 0.51 cm, which is at least 80.23% lower than the other six VLP methods. Experimental results show that when the random selection rate is 50%, the average positioning error of the proposed VLP method is 3.11 cm, which is at least 41.87% lower than the other six VLP methods. The proposed VLP method can be applied to positioning scenarios with different location fingerprinting distributions. The research results can provide a new research idea for the fingerprinting weight calculation method.

Index Terms—Indoor positioning system, location fingerprinting, machine learning, meta-heuristic, visible light positioning.

I. INTRODUCTION

DIFFERENT from outdoor positioning system, indoor positioning system (IPS) mainly uses indoor wireless communication technology to achieve target positioning, such as,

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WiFi [1], Bluetooth [2], radio frequency identification (RFID) [3], wireless sensor network (WSN) [4], etc. In recent years, IPS based on visible light communication (VLC) has become an important candidate for high precision positioning [5], [6], [7]. Compared with radio frequency communication technology, VLC has the following advantages: 1) VLC has no electromagnetic wave radiation; 2) Compared with WiFi, the density of light emitting diode (LED) is higher, and line-of-sight (LOS) links are used for communication between LED transmitter and receiver; 3) LED-based indoor positioning can achieve cm-level positioning accuracy; 4) VLC can realize high-speed wireless data transmission; 5) LEDs have low energy consumption and long lifetime.

Commonly used indoor positioning methods mainly include received signal strength (RSS), angle of arrival (AOA), and time of arrival (TOA), etc. [7], [8]. The RSS value is easily read by the terminal device, which has become one of the hot areas of IPS research. The distance between the target and the LED can be obtained according to Lambertian light source model, and then the positioning of the target can be achieved by least linear square (LLS) and newton-rapson (NR) methods [9], [10], [11], etc. Since the LLS and NR methods are easy to get stuck in the local optimal solution, some authors have combined meta-heuristics to achieve distance-based visible light positioning (VLP). For example, particle swarm optimization (PSO) [12], [13], [14], differential evolution (DE) [15], genetic algorithm (GA) [16], bat algorithm (BA) [17], and grey wolf optimizer (GWO) [18], etc. All of the above VLP methods based on meta-heuristics have two things in common, i.e., the distance between the transceiver needs to be measured and they all rely on a high-precision Lambertian light source model. The VLP methods based on RSS location fingerprinting do not rely on the Lambertian light source model, which makes this approach more favorable [19].

VLP based on fingerprinting mainly uses machine learning, for example, extreme learning machine (ELM), random forest (RF), artificial neural network (ANN), and K -nearest neighbors (KNN), etc. [19], [20]. Among them, KNN is the most commonly used fingerprinting positioning method [21]. If the weights of K nearest neighbor points are different, it is called the weighted KNN (WKNN) [22]. Because the traditional WKNN uses a fixed K value, resulting in a large positioning error (PE). In [23], we proposed an adaptive residual WKNN (ARWKNN) algorithm. A weighted optimum KNN (WOKNN) algorithm was proposed in [24]. In [25], the authors propose a

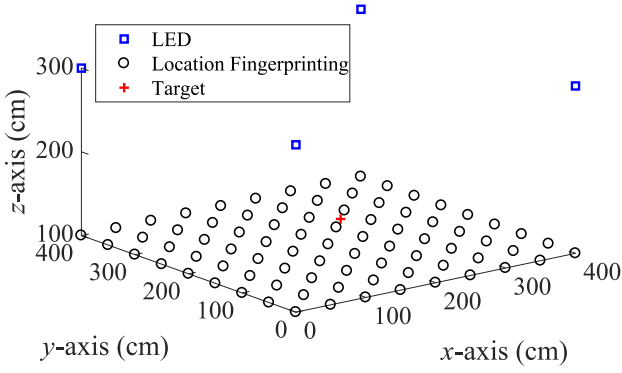


Fig. 1. The VLP model based on fingerprinting method.

fingerprinting localization method using reweighted l_1 -minimization (RW l_1 M) algorithm. A VLP method based on orthogonal matching pursuit (OMP) was proposed in [26]. All of the above fingerprinting-based localization methods have one thing in common, that is, the calculation method of fingerprinting weight generally adopts the mean method or the reciprocal of RSS distance. As a result, the accuracy of the weight of the location fingerprinting is still not high, resulting in a large PE in the case of low-density and non-uniform location fingerprinting distribution.

In recent years, the multi-layer perceptron (MLP) has also been successfully applied to VLP based on fingerprinting method. For example, long short term memory-fully connected network [27], residual concatenation neural network [28], and convolutional neural network [29]. The fingerprinting weight solution method based on MLP has high accuracy but requires a lot of training data [30]. In [30] and [31], the authors use interpolation to reduce the data to be collected. In [32], the authors demonstrate that the VLP system implemented with DIALux software can reduce the data to be collected. In [22] and [33], the authors demonstrate that the use of optical propagation model can effectively reduce the data to be collected, but needs to train a high-precision optical propagation model.

Compared with the existing VLP methods based on location fingerprinting, the main contributions of this article are as follows.

- An accurate VLP method based on location fingerprinting and meta-heuristic is proposed. As far as the authors are aware, this is the first time to report fingerprinting weight calculation method based on meta-heuristic algorithm.
- 15 established meta-heuristics have been compared based on solving the objective function proposed in this paper. To the authors' knowledge, this is the first work in which these meta-heuristics have been applied to solve this problem.
- Under the premise of achieving high positioning precision, there is no need for uniform and high-density location fingerprinting distribution. In order to verify the effectiveness of the proposed VLP method, all possible cases of uneven distribution of location fingerprinting are considered by introducing the random selection rate.
- The effectiveness of the proposed VLP method is compared with six other VLP methods by simulation and experiment.

The research results can provide a new research idea for the fingerprinting weight calculation method.

The remainder of the paper is organized as follows. Section II describes the design of accurate VLP method based on location fingerprinting and meta-heuristic. Simulation results are presented in Section III. Experimental results are presented in Section IV. Finally, Section V concludes the paper.

Notation: Matrices and vectors are in boldface. The transpose operation is denoted by $[\cdot]^T$. The field of real numbers is denoted by $\mathbb{R}$. $\|\cdot\|_2$ denotes the 2-norm of a vector. $\lceil \cdot \rceil$ denotes the rounding up operator.

II. DESIGN OF ACCURATE VLP METHOD BASED ON LOCATION FINGERPRINTING AND META-HEURISTIC

A. Fingerprinting Positioning Model

The VLP model based on fingerprinting method is shown in Fig. 1. Assume that M LEDs are installed on the ceiling. The positioning area is evenly divided into a square grid structure, and each grid point is called a fingerprinting point. The total number of fingerprinting points is N . The grid spacing, i.e., the minimum distance between fingerprinting points is represented by S . It should be noted that in order to simulate the localization performance under non-uniform and low-density location fingerprinting distribution, the random selection rate R_{SR} is introduced. Given the random selection rate, the method of randomly selecting fingerprinting points is adopted, and the number of randomly selected fingerprinting points is $N \cdot R_{SR}$. Due to the random selection of fingerprinting points, the localization performance in all possible non-uniform distribution scenarios can be considered.

B. Measurement Matrix

In the offline phase, the target is placed at the designated N fingerprinting points, and the RSS values from M LEDs are collected, respectively. The measurement matrix $\Phi \in \mathbb{R}^{M \times N}$ is given by

$$\Phi = [\varphi_1 \varphi_2 \cdots \varphi_N] = \begin{bmatrix} \varphi_{1,1} & \varphi_{1,2} & \cdots & \varphi_{1,N} \\ \varphi_{2,1} & \varphi_{2,2} & \cdots & \varphi_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_{M,1} & \varphi_{M,2} & \cdots & \varphi_{M,N} \end{bmatrix}, \quad (1)$$

where the vector φ_n represents the RSS measurement vector of M LEDs collected at the n th fingerprinting point, and is given by

$$\varphi_n = [\varphi_{1,n} \varphi_{2,n} \cdots \varphi_{M,n}]^T. \quad (2)$$

where $n = 1, 2, \dots, N$.

C. Observation Vector

In the online phase, the RSS observation vector $\mathbf{Y}_l \in \mathbb{R}^{M \times 1}$ of M LEDs collected by the l th target at the unknown position Ψ_l is given by

$$\mathbf{Y}_l = [\varphi_{1,l} \varphi_{2,l} \cdots \varphi_{M,l}]^T. \quad (3)$$

where $l = 1, 2, \dots, L$, and L represents the number of targets.

D. Problem Formulation

The PE based on the location fingerprinting is greatly affected by the fingerprinting point weight, so the key is how to calculate the weight of the location fingerprinting. If we can reveal the mathematical essence of the location fingerprinting weight calculation and transform the fingerprinting weight calculation problem into the optimization problem of the objective function, then the global optimization ability of meta-heuristics is used to solve the constrained optimization problem, and the global optimal solution of the location fingerprinting weight is obtained, so as to improve the location accuracy. For this purpose, we established the following objective function.

$$f_1(\mathbf{X}_1) = (\mathbf{Y}_l - \Phi \mathbf{X}_1)^T (\mathbf{Y}_l - \Phi \mathbf{X}_1), \quad (4)$$

where $\mathbf{X}_1 = [g_1, g_2, \dots, g_N]^T$, which represent the weights of N fingerprinting points. Like most fingerprinting weight calculation methods, the value range of g_n is limited to 0 to 1.

It can be seen from problem (4) that when the objective function $f_1(\mathbf{X}_1)$ takes the minimum value, the vector $\mathbf{X}_1$ is the optimal solution. Therefore, the calculation problem of fingerprinting weight is transformed into the optimization problem of objective function. Unfortunately, in (4), since the weights of N fingerprinting points need to be solved, the complexity of the objective function is high, so the objective function needs to be reduced in dimensionality. Since the number of fingerprinting points closest to the target is very small, i.e., most of the elements in the vector $\mathbf{X}_1$ are zero, the key problem of fingerprinting location is to find the K fingerprinting points closest to the target, and solve the weights between the target and K fingerprinting points. The RSS Euclidean distance between the l th target and n th fingerprinting point is given by

$$dis_{l,n} = \left(\sum_{m=1}^M |\varphi_{m,l} - \varphi_{m,n}|^2 \right)^{1/2}, \quad (5)$$

Based on the calculated RSS Euclidean distance, the K fingerprinting points have smallest Euclidean distances can be obtained. According to the obtained K nearest fingerprinting points, a new RSS measurement matrix $\mathbf{H} \in \mathbb{R}^{M \times K}$ is re-established. The objective function is redefined as

$$f_2(\mathbf{X}_2) = (\mathbf{Y}_l - \mathbf{H} \mathbf{X}_2)^T (\mathbf{Y}_l - \mathbf{H} \mathbf{X}_2). \quad (6)$$

where $\mathbf{X}_2 = [g_1, g_2, \dots, g_K]^T$, which represent the weights of K fingerprinting points closest to the target.

In (6), we only need to calculate the weights between the target and K nearest neighbor points. K is much smaller than N , thus greatly reducing the complexity of the objective function. However, due to the nonlinearity of the Lambertian light source model [13], RSS values do not show a simple linear relationship with distance between transceiver, which leads to the problem (6) being a non-convex optimization problem. As we know, the meta-heuristic has strong global optimization ability, so in order to obtain the optimal solution, the meta-heuristic was introduced to solve problem (6).

E. Meta-Heuristics

In general, most meta-heuristics can deal with optimization problems with bound constraints. However, as we know, there is no meta-heuristic that can show obvious advantages for all optimization problems [34]. Aiming at the VLP problem mentioned in this paper, in order to find the optimal solution, 15 established meta-heuristics are used to solve the objective function $f_2(\mathbf{X}_2)$. The basic principles of 15 meta-heuristics are briefly described below.

1) *Biogeography-Based Optimization (BBO)* [35]: The BBO algorithm uses the habitat suitability index (HSI) to indicate the pros and cons of solutions. The higher the HSI, the more species there are in the habitats. The BBO algorithm maintains the balance of the population through the immigration rate and emigration rate. The higher the HSI, the lower the immigration rate and the higher the emigration rate. In order to increase the diversity of the population, the BBO algorithm also introduces a mutation operation.

2) *DE* [36]: The DE strategy uses mutation, crossover, and selection operations to simulate the biological evolution process, and by retaining individuals with good fitness values to find the optimal solution. In the DE strategy, scaling factor and crossover rate are two key parameters. A new candidate solution is created by adding the weighted difference of two different solutions to the other solution in the same population. If the fitness value of the candidate solution is better than the parent, the candidate solution will replace the parent solution in the next generation.

3) *PSO* [37]: In the PSO algorithm, each particle will adjust its flight speed and flight trajectory according to the individual optimal solution and the global optimal solution. The flying speed of each particle is affected by the inertia weight, cognitive learning factor and the social learning factor. The cognitive learning part represents the learning experience of the individual, and the social learning part represents the learning experience of the population.

4) *Artificial Bee Colony (ABC)* [38]: The ABC algorithm includes three types of bees: employed, onlooker and scout bees. The employed and onlooker bees explore within a single neighborhood. The scout bees perform random searches in the entire solution space. Each employed bee has a definite food source, and the employed bees will feed back the food source information to the onlooker bees. The candidate solution is represented by the location of the food source, and the amount of honey in the food source represents the fitness value of the corresponding candidate solution.

5) *Evolution Strategy With Covariance Matrix Adaptation (CMA-ES)* [39]: One of its innovations is de-randomization, the new candidate solution is selected according to the multi-variate normal distribution. The core idea is to adaptively adjust the covariance matrix of the multi-dimensional normal search distribution, so that the population converges to the optimal solution. The other core idea is the cumulative evolutionary path, CMA-ES updates the population solutions according to the step length and covariance matrix, according to the fitness value of the solution vector, the step length and covariance matrix are updated by the evolution path.

6) *GWO* [40]: There are four types of gray wolves in the GWO, which are called alpha, beta, delta, and omega. In the population, alpha has the highest rank, beta has the second highest rank, delta has the third rank, and omega is the lowest ranking wolf, playing the role of a follower. For optimization problems, alpha represents the optimal solution, the second and third best solution is named beta and delta, respectively, and omega represents other candidate solutions. The position of the prey is determined by alpha, beta, and delta, the remaining gray wolves randomly generate a feasible solution around the position of the prey.

7) *BA* [41]: Bats can usually use echolocation to perceive the distance to the prey. Each bat will update its flight trajectory according to its frequency, wavelength and loudness, and the frequency or wavelength will vary with the distance between the bat and the prey. The loudness is assumed to vary from a large value to a minimum value. The position of the bat in the search space corresponds to the candidate solution of the objective function. A random walk is applied for generate a new solution for each bat. The update of the speed and position of the BA is similar to the standard PSO.

8) *Cultural Algorithms (CA)* [42]: The key to CA is how to define population space and belief space. The population space is used to implement any population-based evolutionary algorithm. In the population space, by evaluating the fitness value of each individual, some individuals can be replaced by some of its offspring. The belief space selects individuals from the population space through the acceptance functions. The current state of the belief space can be used to modify the performance of individuals in the population. In order to describe the mutual influence mechanism between the current belief space and population space, a communication protocol between the two spaces is established.

9) *Invasive Weed Optimization (IWO)* [43]: In IWO algorithm, the number of seeds produced by a plant depends on its fitness, the minimum fitness and the maximum fitness of the population. The better the fitness, the more seeds will be produced. The seeds produced by plants are randomly distributed in the feasible solution space with a normal distribution. The standard deviation will decrease nonlinearly as the number of iterations increases, in order to enhance the local search ability of the population in the later stage. If the number of solutions is exceeded, according to the competitive survival law, individuals with good fitness can survive.

10) *Firefly Algorithm (FA)* [44]: The changing law of light intensity and attractiveness are two key issues of the FA, attractiveness is proportional to brightness, so for any flashing fireflies, the less bright fireflies will move closer to the brighter fireflies. If the brightness is the same, the fireflies move randomly. For the optimization function, the light intensity depends on the fitness value of the location of the firefly, the higher the brightness, the better the location, i.e., the better the fitness value, and the search optimization process was simulated as the attraction and movement process of individual fireflies.

11) *Teaching-Learning-Based Optimization (TLBO)* [45]: In the TLBO, the population is considered as a group of learners. The population in the TLBO method will be updated by two

stages including teacher and learner phases. In the teacher phase, through learning from teachers, keep narrowing the distance between learners and teachers. In the learner phase, through mutual learning between learners, a learner with poor fitness value will move to a learner with better fitness value. When solving the objective function, the learner is regarded as the candidate solution, and the teacher is regarded as the optimal solution obtained so far.

12) *Cuckoo Search (CS)* [46]: In CS algorithm, first, randomly selects a nest for storage the egg laid by the cuckoo. Second, in a random selection of nests, the best nests will be reserved for the next generations. Last, assuming that the number of host nests is constant, and the eggs laid by cuckoos will be found by the host bird with a certain probability. In each iteration process, by evaluating the pros and cons of feasible solutions, the better solutions are retained, and the same number of new solutions as the discarded solutions are generated according to the random walk method.

13) *Bees Algorithm (BSA)* [47]: In the BSA, first, the scout bees move randomly in the search space. The position of the bee is the position of the food, and it is expressed by the solution of the optimization problem. Second, assess the pros and cons of the site visited by the scout bees. Third, the bees with the best fitness will be selected as "selected bees". Then, assigns more bees to search near the best e sites. Last, the remaining bees move randomly in the search space to find new feasible solutions.

14) *GA* [48], [49]: In the standard GA, the three operations of selection, crossover and mutation are executed orderly in each iteration. The selection operator controls the direction of the search. Selection operation embodies the principle of survival of the fittest, selecting high-quality individuals and eliminating low-quality individuals according to the fitness of the solution. There are two key parameters in GA, namely crossover rate and mutation rate. Through the crossover operation, the offspring inherited the characteristics of the two parent structures. Through the mutation operation, increasing the diversity of the population.

15) *Simulated Annealing (SA)* [50]: First, the SA algorithm needs to give an initial temperature value. Then, the parent mutates in a certain pattern to produce a new offspring, which is called the candidate solution of the optimization problem. For a minimum objective optimization function, if the fitness of the candidate solution is less than that of the parent, then the offspring replace the parent, otherwise the offspring will be accepted with a Boltzmann probability.

F. Exploitation and Exploration Mechanisms

The meta-heuristic algorithm has two very important stages, i.e., local exploitation and global exploration. The global exploration stage refers to the process of investigating the promising areas in the search space as widely as possible, while the local exploitation stage refers to the local search capability around the potential area obtained in the exploration stage. Generally, for a non-convex optimization problem, the stronger the local exploitation ability is, the faster the algorithm converges, but the

TABLE I
LOCAL EXPLOITATION AND GLOBAL EXPLORATION MECHANISM FOR
DIFFERENT META-HEURISTICS

Meta-heuristics	Local exploitation	Global exploration
PSO	Last flight speed and position	Cognitive learning and social learning
DE	Mutation	Crossover
SA	Annealing criterion	Metropolis criterion
ABC	Local search by employed and onlooker bees	Global search by scout bees
GWO	Attacking prey	Search for prey
GA	Mutation	Crossover
FA	Light intensity and attractiveness	Best firefly
CMA-ES	Evolution path	Adaptive adjustment of covariance matrix
CA	Belief space	Population space
IWO	Competitive exclusion	Spatial dispersal
BBO	Mutation operator	immigration rate and emigration rate
TLBO	Learning phase	Teaching phase
BA	Low loudness and high pulse emission rate	High loudness and low pulse emission rate
CS	Levy flight	Global best solutions
BSA	Select sites for neighbourhood search	The fittest bee was selected

easier the population is to fall into the local optimal solution. On the contrary, the stronger the global exploration ability, the slower the convergence speed of the algorithm, but the population is not easy to fall into the local optimal solution. How to find a balance between exploitation and exploration is the research difficulty of meta-heuristic, which is why no meta-heuristic can perform optimally on all optimization problems so far [34], [51]. For some specific problems, some meta-heuristics can show good global optimization ability. On the contrary, some meta-heuristics are more likely to fall into local optimal solutions. This is why this paper investigates 15 meta-heuristics in detail for the fingerprinting weight optimization problem of VLP. Local exploitation and global exploration mechanism for different meta-heuristics are shown in Table I.

A very important issue is how to apply meta-heuristics to solve problem (6), which is the weight optimization problem of location fingerprinting proposed in this paper. In the weight optimization problem of location fingerprinting, the variable is the weight value of location fingerprinting, and the dimension represents the number of nearest points. Therefore, it is necessary to discretize the weight value. According to the weight optimization function of location fingerprinting and the iteration rules of meta-heuristics, the optimal solution of location fingerprinting weight can be found through the cooperation between populations.

Algorithm 1: Accurate VLP Method Based on Location Fingerprinting and Meta-Heuristic.

Input: $K, \Phi \in \mathbb{R}^{M \times N}$, and $Y_l \in \mathbb{R}^{M \times 1}$.

Output: The coordinate Ψ_l of the l th target.

Step 1: Calculate the RSS Euclidean distance according to (5).

Step 2: Sort vector dis ,

$[A, I] = \text{sort}(dis)$.

where $dis = [dis_{l,1}, dis_{l,2}, \dots, dis_{l,N}]^T \in \mathbb{R}^N$, $\text{sort}(dis)$ means to sort the vector dis in ascending order. $A \in \mathbb{R}^N$ and $I \in \mathbb{R}^N$ represent the sorted distance vector and the corresponding index set, respectively.

Step 3: According to K nearest neighbor points, build matrix $H \in \mathbb{R}^{M \times K}$.

for $i = 1: K$

$H(:, i) = \Phi(:, I(i))$;

end for

Step 4: Calculate the weights of K nearest neighbor points.

Step 4.1: Initialize the parameters of the meta-heuristic.

Step 4.2: Establish an objective function based on matching RSS residuals according to (6).

Step 4.3: Output the weights of K nearest neighbor points.

while (stopping criterion is not met)

do

Meta-heuristic algorithm.

end while

Output: $X_2 = \arg_{X_2} \min(f_2(X_2))$; s.t. $0 < g_k < 1$.

Step 5: Compute Ψ_l ,

$\Psi_l = \Omega X_2$.

where $\Omega \in \mathbb{R}^{2 \times K}$ represents the 2-D coordinate matrix of the K nearest neighbor points.

G. Algorithm Design

The pseudocode of the proposed VLP method as shown in Algorithm 1. First, the calculation problem of fingerprinting weight is transformed into the optimization problem of objective function. Second, 15 established meta-heuristics are used to solve the objective function. The fingerprinting weight vector X_2 is obtained by solving the objective function $f_2(X_2)$.

It can be seen from the simulation results that the positioning performance based on the DE algorithm is the best. Due to the limitation of the length of the paper, this paper only introduces the solving process of the objective function based on the DE. For a detailed introduction to other meta-heuristics, please refer to the relevant literature. The pseudocode of solving the objective function proposed in this paper with DE is shown in Algorithm 2. In Algorithm 2, t represents the current number of iterations. CR stands for crossover probability. F stands for differential scaling factor. $\text{randb}(j)$ indicates that a value is randomly generated between $[0, 1]$. $\text{rnbr}(i)$ represents a random number in $[1, 2, \dots]$.

Algorithm 2: Solving the Problem (6) by DE Algorithm, i.e., Step 4 in Algorithm 1.

Step 1: Initialize the population size N_P , and the initial random solution vector $\mathbf{x}_i(t)$ corresponding to each individual. The fitness value of each individual solution vector is evaluated according to (6).

Step 2:
while (stopping criterion is not met)
do
for $i = 1 : 1 : N_P$
 for $j = 1 : 1 : K$
Step 2.1: Mutation operation
 $v_{i,j}(t+1) = x_{r_1,j}(t) + F(x_{r_2,j}(t) - x_{r_3,j}(t))$
where the index values r_1, r_2 , and r_3 are three values randomly generated between $[1, N_P]$, and need to satisfy $r_1 \neq r_2 \neq r_3 \neq i$.
Step 2.2: Crossover operation
 if ($\text{randb}(j) \leq CR$ or $j = \text{rnbr}(i)$)
 then $u_{i,j}(t+1) = v_{i,j}(t+1)$
 else $u_{i,j}(t+1) = x_{i,j}(t)$
 end if
end for
Step 2.3: Selection operation
 if $f_2(\mathbf{u}_i(t)) \leq f_2(\mathbf{x}_i(t))$
 then $\mathbf{x}_i(t+1) = \mathbf{u}_i(t)$
 else $\mathbf{x}_i(t+1) = \mathbf{x}_i(t)$
 end if
end for
end while

K . $\mathbf{x}_i(t)$ represents the weight vector of K nearest neighbor points of the i th particle in the t th iteration, i.e., $\mathbf{x}_i(t) = [x_{i,1}(t), x_{i,2}(t), \dots, x_{i,K}(t)]^T = \mathbf{X}_2 = [g_1, g_2, \dots, g_K]^T$.

III. SIMULATION ANALYSIS

In order to verify the positioning performance of the proposed VLP method, we consider the following VLP methods based on location fingerprinting for comparison: WKNN based on Euclidean distance (ED) metric method [22], i.e., WKNN-ED; WKNN based on square chord distance (SCD) metric method [22], i.e., WKNN-SCD; multi-layer perceptron (MLP) [30]. linear-regression machine learning (LRML) [32]. Fingerprinting location methods based on compression sensing include RWI₁M [25] and OMP [26].

A. Measurement Model

Without loss of generality, in the simulation system, we only consider the Lambertian light source model of the LOS communication, which are widely adopted by many VLP papers such as [22], [23], [24], [25]. The light power value received by the target from the m th LED is

$$P_r = P_t \frac{A(b+1)T_s g}{2\pi d_m^2} (\cos(\lambda_m))^b \cos(w_m). \quad (7)$$

where $m = 1, 2, \dots, M$. P_r represents the light power received by the target; P_t represents the transmitted optical power; g represents optical concentrator gain; b is the Lambertian order; d_m is the distance between the sender and the receiver; T_s represents optical filter gain; A is the detector physical area; w_m and λ_m are the incident and irradiance angles, respectively.

B. Positioning Performance Evaluation

The PE E_l Is Given By

$$E_l = \|\Psi_l - \Psi'_l\|_2, \quad (8)$$

where Ψ'_l represents the actual location of the target.

The Average PE E_{APE} Is Given By

$$E_{APE} = \frac{1}{L} \sum_{l=1}^L E_l, \quad (9)$$

The Standard Deviation E_{STD} of the PE Is Given By

$$E_{STD} = \sqrt{\frac{1}{L} \sum_{l=1}^L (E_{APE} - E_l)^2}. \quad (10)$$

C. Simulation Parameters

The parameters setting of the Lambertian light source model obeys the typical LED parameters setting, as follows: $P_t = 6$ W, $T_s = g = 1$, $A = 1$ cm², $b = 1$. $\cos(\lambda_m) = \cos(w_m) = h_m/d_m$, h_m represents the vertical distance between the sender and the receiver, which are widely adopted in papers such as [14], [15], [17], [22], [23], [25]. 4 LEDs are installed on the ceiling. The size of positioning space is 4 m × 4 m × 3 m. $L = 200$. Unless otherwise specified, the sparsity of OMP and RWI₁M algorithms is 4, i.e., $K = 4$, which shows the best positioning performance [22], [23], [24], [25]. The MLP architecture is the same as in [30], including input layer, hidden layer, and output layer. The hidden layer is a single layer structure. The number of neurons in the input layer is 4, indicating the RSS values from the 4 LEDs. The number of neurons in the output layer is 2, which is used to calculate the 2-D coordinate. After comprehensive evaluation of the positioning accuracy, the optimal number of hidden nodes is selected to be 15. LRML is no need to set parameters [32]. The simulation tool is MATLAB R2017a.

In this paper, 15 established meta-heuristics have been compared based on solving $f_2(\mathbf{X}_2)$, i.e., problem (6), and the parameters of 15 established meta-heuristics are set as follows: BBO, the habitat modification probability is 0.9, immigration probability bounds per gene = [01], the maximum migration rate is 1, and mutation probability is 0.1; DE/rand/1 strategy is used. The scaling factor is 0.3, and the crossover rate is 0.6; PSO, the inertia weight is defined as a linear decrease instead of a fixed constant as the number of iterations increases, the max inertia weight and the min inertia weight are 0.9 and 0.4, respectively. Cognitive learning factor and social learning factor are both equal to 1.5; ABC, the number of food sources, onlooker bees and employed bees are all equal to half of the population size; There is no other control parameters need to be set in CMA-ES, GWO and TLBO; BA, the pulse rates and loudness are 0.1 and 0.6, respectively, the maximum and minimum frequency values

TABLE II
COMPARATIVE PERFORMANCE, BEST VALUES ARE HIGHLIGHTED IN BOLD

Meta-heuristics	E_{APE} (cm)	E_{STD} (cm)	$P(E_l \leq 5)$
PSO	2.07	1.69	0.95
DE	1.86	1.43	0.97
SA	2.11	1.63	0.94
ABC	3.89	2.28	0.75
GWO	2.39	1.98	0.91
GA	3.62	2.56	0.77
FA	2.17	1.80	0.91
CMA-ES	4.72	3.32	0.63
CA	2.25	2.01	0.93
IWO	2.31	2.22	0.89
BBO	3.19	2.58	0.79
TLBO	2.23	1.70	0.91
BA	4.80	3.74	0.69
CS	2.05	1.51	0.95
BSA	4.07	2.59	0.71

are 1 and 0, respectively, and the initial emission rate is 0.1; CA, the acceptance rate is 0.35. IWO, the initial population is 10. Nonlinear modulation index is 2. The maximum and minimum number of seeds are 5 and 0, respectively. The initial and final values of the standard deviation are 0.5 and 0.001, respectively; FA, light absorption coefficient is 1, and the initial attractiveness is 2; CS, the step size is 1, and the probability of discovery by the host bird is 0.2; BSA, the number of bees recruited for best sites, selected sites, bees recruited for selected sites, and selected elite sites are 50, 25, 25, and 10, respectively. The initial patch size is 0.1; GA, the crossover rate and mutation rate are 0.7 and 0.3, respectively; SA, the initial temperature is 0.1, and the temperature reduction rate is 0.99.

It is worth noted that the parameters used for each meta-heuristic are adjusted to have appropriate values in order to obtain the best positioning performance, including the population size and number of iterations, but we did not make any changes to the 15 meta-heuristics. Unless otherwise specified, the population size and maximum number of iterations are set to 100 and 200, respectively. In Step 4.3 of algorithm 1, the termination conditions are set to reach the maximum number of iterations or meet the RSS reconstruction accuracy requirements, and the reconstruction accuracy is set to 10^{-3} .

D. Result Analysis

When signal-to-noise ratio (SNR) is 20 dB, and S is 50 cm, the positioning performance of 15 different meta-heuristics is shown in Table II. $P(E_l \leq 5)$ is the probability that the PE is less than or equal to 5 cm. Obviously, the higher the $P(E_l \leq 5)$ value, the smaller the E_{APE} and E_{STD} values, the better the positioning performance. Table II shows that most meta-heuristics have comparable positioning performance. Table II also show that DE not only provides the smallest E_{APE} and E_{STD} values, but also provides the largest $P(E_l \leq 5)$ value.

As described in Section II-F, 15 meta-heuristics have different exploitation and exploration mechanisms. For the problem (6), ABC, CMA-ES and BSA have strong local exploitation

ability, but it is easy to fall into local optimization. PSO can better balance local exploitation and global exploration through cognitive learning and social learning mechanisms. TLBO can also better balance local exploitation and global exploration through learning and teaching mechanisms. CA can also better balance local exploitation and global exploration through belief space and population space. Compared with PSO and TLBO, BA updates the iteration of population mainly through social learning mechanism, lacking self-learning mechanism. When solving problem (6), BA does not achieve a good balance between local exploitation and global exploration. Compared with the BSA, although CS also relies on the global optimal solution, the Levy flight mechanism of CS algorithm is also very suitable for solving problem (6). FA can balance between local exploitation and global exploration by simulating the light intensity change and formulating attractive rules. GWO also depends on the global optimal position, i.e., alpha, to some extent, but also depends on the second and third optimal solutions, i.e., beta and delta. Although the location update of gray wolves does not depend on the historical optimal location of individuals, the location of prey is determined by alpha, beta, and delta. Therefore, compared with BA, it can achieve better positioning performance. IWO maintains balance between local exploitation and global exploration by controlling the standard deviation of the normal distribution. Its standard deviation will decrease nonlinearly with the increase of the number of iterations, thus ensuring that the algorithm has better global exploration capability at the initial stage and better local exploitation capability at the later stage.

Although both DE and GA use mutation and crossover mechanisms to balance between local exploitation and global exploration, DE obviously achieves better positioning performance. The difference between DE and GA is that the mutation vector of DE works by adding the weighted difference of two different solutions to the other solution in the same population, thereby creating a new candidate solution and preserving the candidate solution by greedy selection. The mutation vector of the GA controls the parent crossing according to the fitness value. The individual with the best fitness value will have a higher probability of being selected, which makes it easier to lose the diversity of the population. In order to increase the diversity of the population, the BBO also introduces a mutation operation, which makes both low HSI solutions and high HSI solutions are more prone to mutate. When solving problem (6), it is easy to cause the population to jump out of the global optimal solution. The metropolis criterion mechanism of SA is also very suitable for solving problem (6). In conclusion, as DE provides the best positioning performance among the 15 meta-heuristics, it will be selected to solve the problem (6) in the following numerical simulations and experiments.

The impact of SNR on the average PEs is shown in Fig. 2. It shows that the average PE of the proposed VLP method is the smallest. WKNN-ED and WKNN-SCD are greatly affected by noise. The weight calculation method of WKNN is still not accurate enough, and the average PE is still large even at a high SNR value. Although the LRML method is less affected by noise, the average PE is larger. Due to the nonlinearity of the

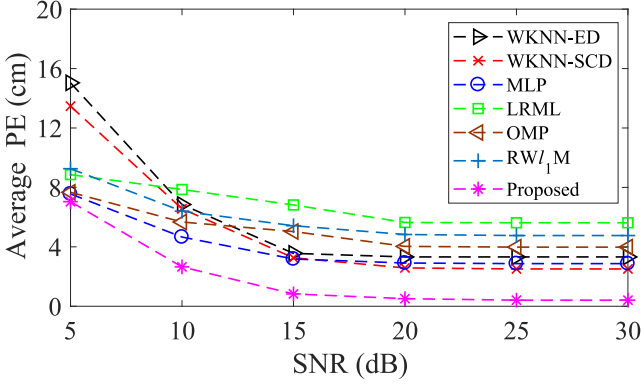
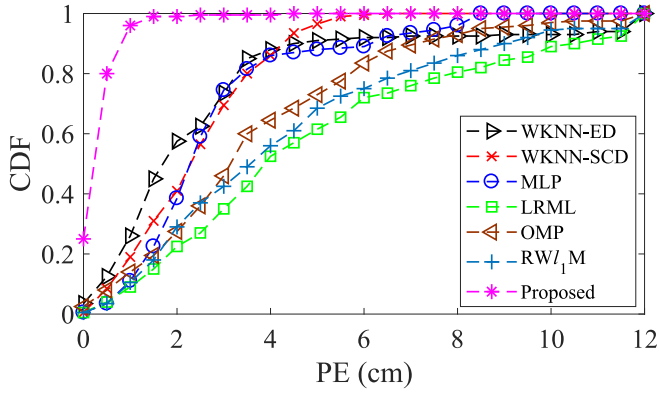
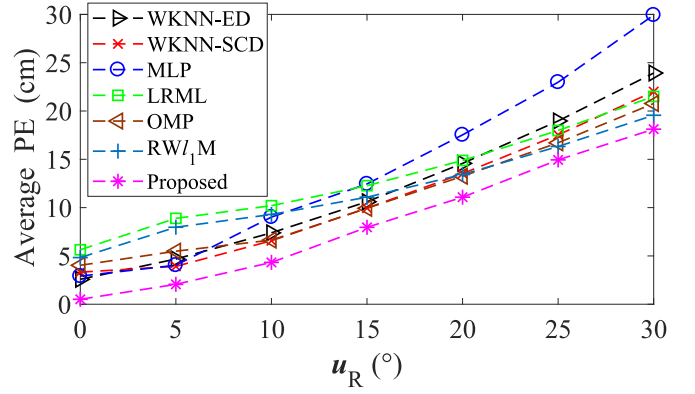
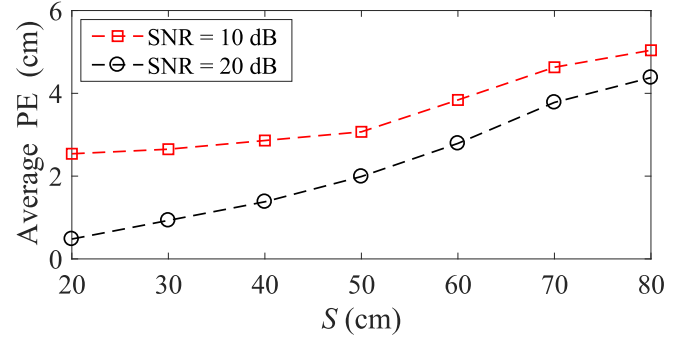

 Fig. 2. The impact of SNR on the average PEs with $S = 20$ cm.


Fig. 3. The cumulative distributions of PEs with SNR = 20 dB.

Lambertian light source model, RSS values do not show a simple linear relationship with distance, which leads to the problem (6) that is a non-convex optimization problem. The average PEs based on OMP and RWl_1M are less affected by noise. However, due to the low density of LED, i.e., M is 4, resulting in insufficient observations of compression reconstruction, and the signal reconstructed by linear least square method is easy to fall into the local optimal solution, resulting in large average PEs of OMP and RWl_1M . The average PE of the proposed VLP method is still very small even under the condition of low SNR value, that is, in the range of 5 dB to 15 dB.

When S is 20 cm, the cumulative distribution functions (CDFs) of PEs are shown in Fig. 3. Fig. 3 shows that the CDF of PE of the proposed VLP method is optimal. The average PEs based on WKNN-ED, WKNN-SCD, MLP, LRML, OMP, RWl_1M , and proposed VLP methods are 3.33 cm, 2.58 cm, 2.97 cm, 5.64 cm, 4.03 cm, 4.84 cm and 0.51 cm, respectively. Compared with WKNN-ED, WKNN-SCD, MLP, LRML, OMP, and RWl_1M methods, the average PE of the proposed VLP method can be reduced by 84.68%, 80.23%, 82.83%, 90.96%, 87.34%, and 89.46%, respectively.

When SNR is 20 dB, the impact of u_R on the average PEs is shown in Fig. 4, where u_R represents the receiving orientation of PD and rotates along the z - x plane. The relationship between orientation u_R and incident angle w_m can be found in [13]. As expected, as the value of u_R increase, the average PE of the


 Fig. 4. The impact of u_R on the average PEs with $S = 20$ cm.

 Fig. 5. The impact of S on the proposed VLP method.

seven different VLP methods will increase. We can see that the proposed VLP method has the smallest impact from the tilt of receiver and the smallest average PE. It is worth noted that when the RSS database is built offline, the orientation u_R remains unchanged, i.e., vertically upward, which also leads to inaccurate parameters obtained by MLP offline training when the tilt angle of PD is large, resulting in large PEs. In order to reduce the impact from the tilt of receiver, RSS values can be collected at different tilt angles of PD in the offline stage. Although the acquisition of RSS values from different tilt angles by the receiver will increase the cost of building RSS database, it may be an effective method to reduce the impact from the tilt of receiver, but this is not discussed in this paper.

The variation of the average PE of the proposed VLP method with the grid spacing S as shown in Fig. 5. As expected, as the S becomes larger, the PE becomes larger. Although the PE becomes smaller as the S decrease, the increase in fingerprinting points will increase the difficulty of building RSS database in the offline. However, even if the grid spacing reaches 80 cm, at a SNR of 10 dB, the average PE of the proposed VLP method is 5.04 cm, which is fully in line with most actual positioning application scenarios.

The average PE of the proposed VLP method as a function of the population size and the number of iterations is shown in Figs. 6 and 7, respectively. As expected, as the population size and the number of iterations increases, the average PE gradually decreases. The time complexity of population based meta-heuristic optimization mainly depends on the population

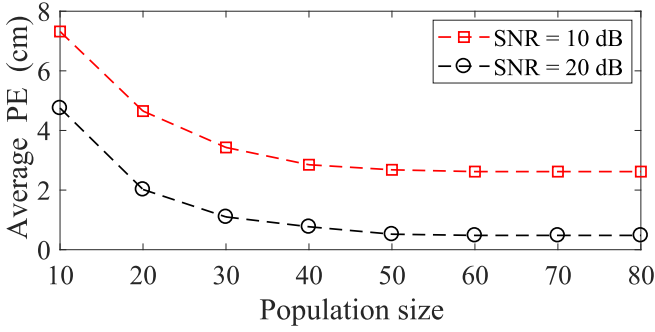


Fig. 6. The impact of population size on the proposed VLP method with $S = 20$ cm.

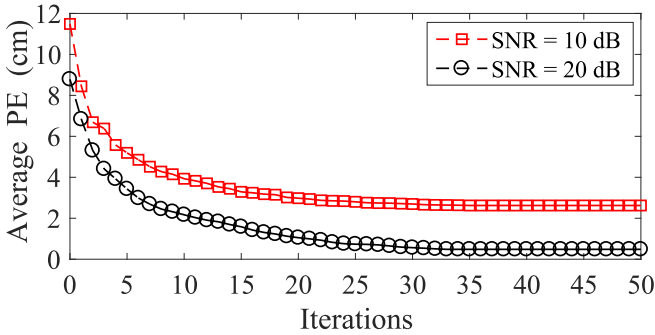


Fig. 7. The impact of the number of iterations on the proposed VLP method with $S = 20$ cm.

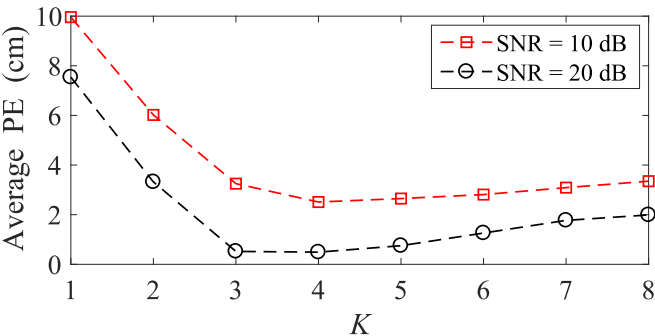


Fig. 8. The impact of K on the proposed VLP method with $S = 20$ cm.

size and the number of iterations. Figs. 6 and 7 show that when the average PE converges, the population size and the number of iterations of the proposed VLP method are small, with values of 50 and 30, respectively, so the time complexity is relatively low.

The variation of the average PE of the proposed VLP method with the number of nearest neighbor fingerprinting points K is shown in Fig. 8. It is shown that the average PE first decreases and then increases. When K is 4, the average PE is the smallest. When K is lower than the optimal value, the positioning information will be lost. When K is too high, fingerprinting points with low correlation will be involved in the positioning, resulting in larger PEs in both cases.

In order to verify that the proposed VLP method still has good positioning performance under non-uniform location

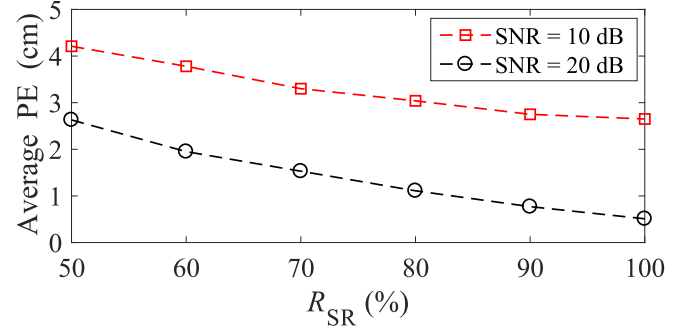


Fig. 9. The impact of R_{SR} on the proposed VLP method with $S = 20$ cm.

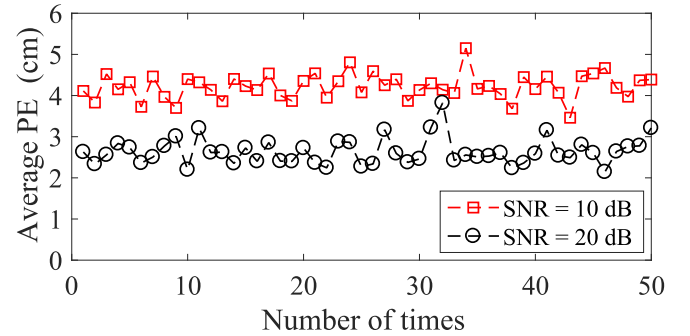


Fig. 10. The variation curve of the average PE of the proposed VLP method with the number of times, $R_{SR} = 50\%$.

fingerprinting distribution, we introduce the influence of random selection rate R_{SR} on the PE. In order to obtain the result more accurately, under the same random selection rate, the average value method of repeated execution 100 times is adopted. Fig. 9 shows that even if the R_{SR} is only 50%, that is, the location fingerprinting distribution density is low and extremely uneven, the average PE is still very small.

We use the standard deviation of PEs to represent the robustness of positioning, that is, the smaller the standard deviation, the better the stability and robustness of positioning results. The positioning results of repeated executions of the proposed VLP method 50 times are shown in Fig. 10. Even if the random selection rate is only 50%, the fluctuation of the average PE curve for 50 repeated executions is still small. When the SNR is 10 dB and 20 dB, the standard deviation is 0.32 cm and 0.29 cm, respectively. Therefore, the proposed VLP method has good robustness.

IV. EXPERIMENTAL ANALYSIS

In order to evaluate the positioning performance of the proposed VLP method in an actual environment, we have established a practical experimental platform for VLP method based on location fingerprinting, this simple and practical VLP experimental platform is mainly composed of PD, op-amp, voltage comparator and microcontroller unit, which has been widely used in papers such as [12], [16], [24], [30], [52], [53], [54]. The experimental scene is shown in Fig. 11, which is mainly composed of 4 LEDs, computer numerical control (CNC) machine and a PIN photodiode (PD) receiver. On the

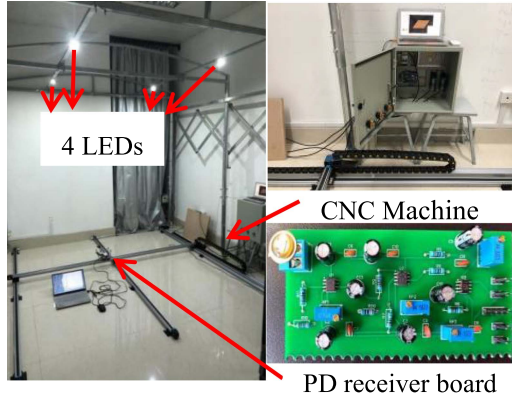


Fig. 11. A practical experimental platform for VLP based on location fingerprints.

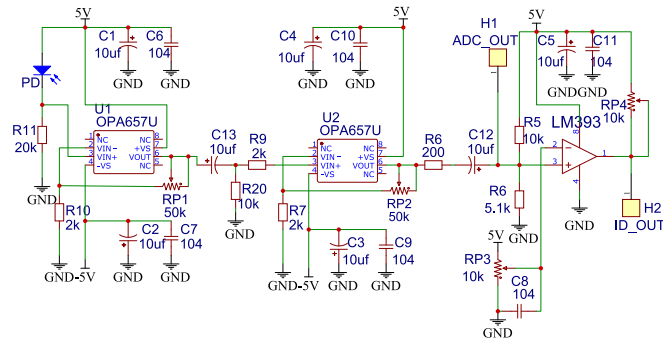


Fig. 12. Schematic diagram of the PD receiving board.



Fig. 13. The coding format of the LED transmission signal.

basis of reference [54], we designed the schematic diagram of PD receiving board, as shown in Fig. 12, LED signals received by PD form two channels after passing through two-stage op-amp circuit, among which one channel is converted by analog-digital converter (ADC) to obtain RSS value, and the other channel is converted by voltage comparator to generate LED identification (ID). The model of microcontroller unit is STM32F407. The vertical distance between the PD and the LED is 1.5 m. The 4 LEDs are evenly distributed on the same horizontal plane, and the 2-D coordinates are (0.5 m, 0.5 m), (1.5 m, 0.5 m), (0.5 m, 1.5 m), and (1.5 m, 1.5 m), respectively. We use time division multiplexing technology to distinguish signals from different LEDs [24]. The four LEDs have synchronization frames, and each LED is assigned five cycles to send signals, as shown in Fig. 13. The starting bit of the signal is sent in T_1 cycle, and the ID code of LED sent in T_2 and T_3 cycles. For the sake of

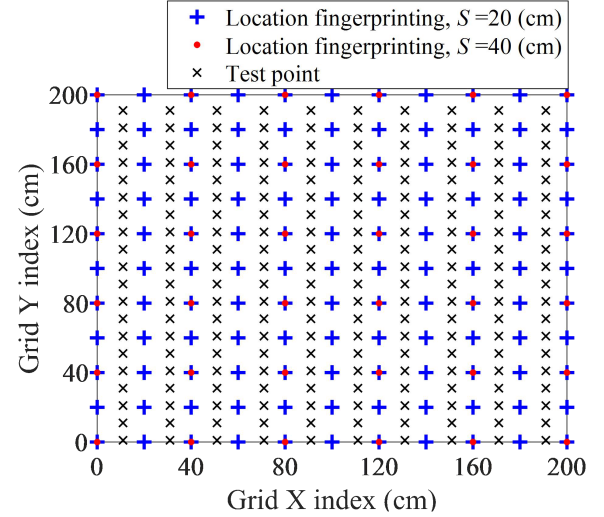
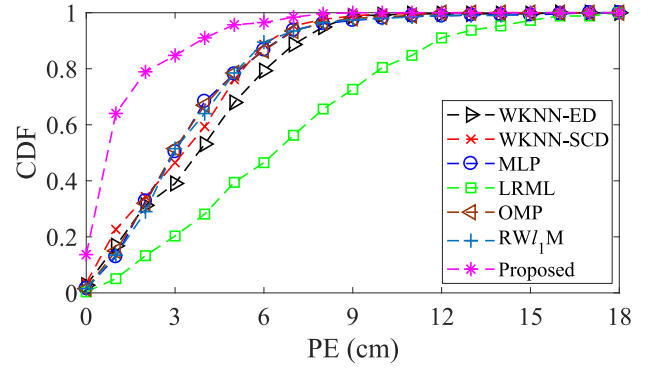


Fig. 14. Distribution map of location fingerprints and test points.


 Fig. 15. The cumulative distributions of PEs with $S = 20$ cm.

simplicity, the ID codes of four LEDs are represented by codes of 00, 01, 10 and 11, respectively, and 0 and 1 are represented by a duty cycle of 20% and 80%, respectively. For example, the ID code of the LED shown in Fig. 13 is 01. T_4 starts to measure the voltage value, and the RSS value is obtained after ADC conversion. T_5 cycle is the stop bit. The actual positions of fingerprinting points and test points were measured by a 2-D CNC machine. The measurement accuracy of the CNC machine is 0.1 mm, which is several orders of magnitude better than the expected positioning accuracy of the VLP system. Therefore, the real coordinates can be recorded automatically and accurately. The PD receiver was installed on the CNC machine to ensure that the orientation angle of the PD receiver remains constant.

The distribution map of location fingerprinting points and test points are shown in Fig. 14. The RSS is collected 100 times at each point, and the average value is calculated. The time of collecting each RSS value depends on the emission rate of LED and the sampling rate of ADC. In Fig. 14, the grid spacing S of the location fingerprinting is divided into two cases, i.e., 20 cm and 40 cm. 200 different positions have been selected in a positioning space to test the PE.

Figs. 15 and 16 show that no matter the grid spacing S is equal to 20 cm or 40 cm, the CDF of PE of the proposed VLP method is

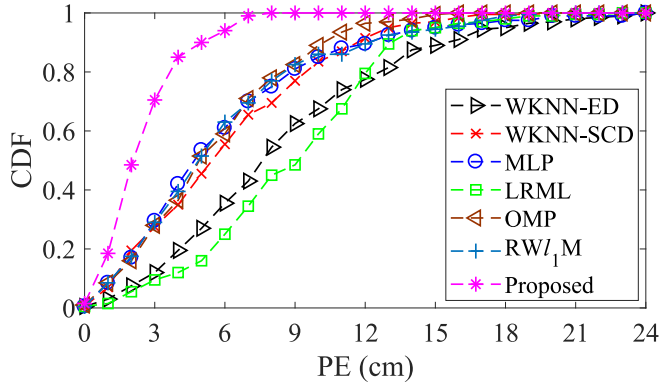


Fig. 16. The cumulative distributions of PEs with $S = 40$ cm.

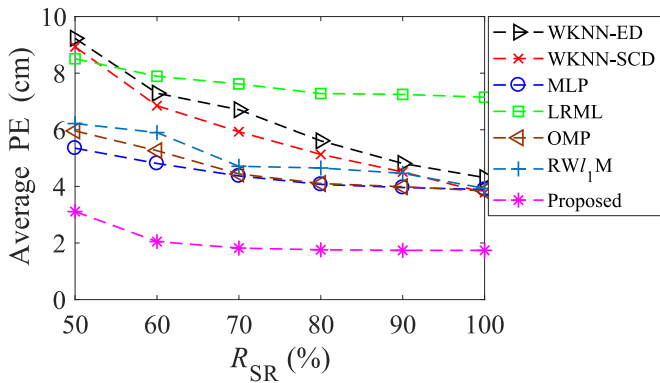


Fig. 17. The impact of $R_{SR}(\%)$ on the average PEs with $S = 20$ cm.

optimal. When S is 40 cm, the average PEs based on WKNN-ED, WKNN-SCD, MLP, LRML, OMP, $RW_{I1}M$, and proposed VLP methods are 8.92 cm, 6.48 cm, 6.39 cm, 9.25 cm, 5.96 cm, 6.24 cm and 2.92 cm, respectively. Compared with WKNN-ED, WKNN-SCD, MLP, LRML, OMP, and $RW_{I1}M$ methods, the average PE of the proposed VLP method can be reduced by 67.26%, 54.94%, 54.30%, 68.43%, 51.01%, and 53.21%, respectively.

At the same R_{SR} value, the results of repeated execution 100 times are shown in Fig. 17. As expected, with the decrease of R_{SR} value, the distribution of fingerprinting positions is more uneven and sparse, and the greater the PE. Fig. 17 shows that the proposed VLP method has the smallest impact from the random selection rate and the smallest average PE. When R_{SR} is 50% and 70%, respectively, the average PEs of five VLP methods are shown in Table III. The results show that the average PE of the proposed VLP method is minimal.

In order to analyze the PEs of 200 test points under different non-uniform fingerprinting distributions, when the random selection rate is 50%, the results of repeated execution 100 times are shown in Fig. 18. When the random selection rate is 50%, we can see from Table III that the average PE of the proposed VLP method is 3.11 cm, compared with WKNN-ED, WKNN-SCD, MLP, LRML, OMP, and $RW_{I1}M$ methods, the average PE of the proposed VLP method can be reduced by 66.31%, 65.17%, 41.87%, 63.45%, 47.73%, and 51.63%, respectively.

TABLE III
THE AVERAGE PE OF FIVE VLP METHODS VARIES WITH THE VALUE OF R_{SR}

Algorithm	Average PE ($R_{SR} = 50\%$)	Average PE ($R_{SR} = 70\%$)
WKNN-ED	9.23 cm	6.71 cm
WKNN-SCD	8.93 cm	5.93 cm
MLP	5.35 cm	4.37 cm
LRML	8.51 cm	7.62 cm
OMP	5.95 cm	4.44 cm
$RW_{I1}M$	6.43 cm	4.71 cm
Proposed	3.11 cm	1.82 cm

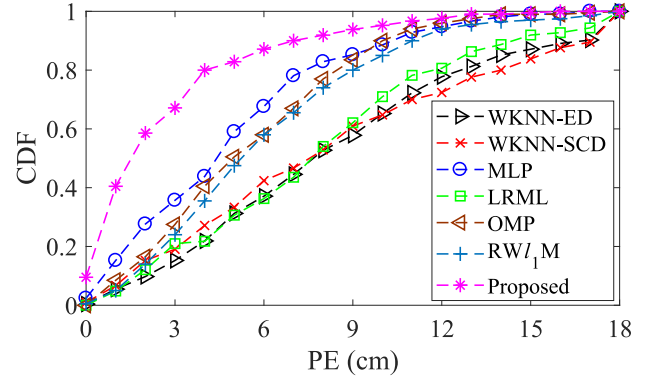


Fig. 18. The cumulative distributions of PEs with $R_{SR} = 50\%$.

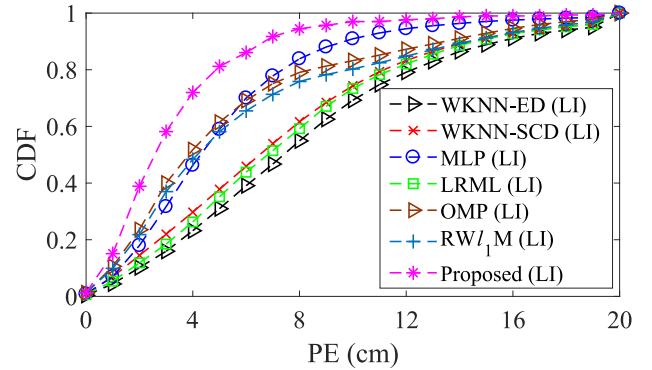


Fig. 19. The cumulative distributions of PEs after linear interpolation with $R_{SR} = 40\%$.

In order to verify that the method of reducing data collection can also be applied to the proposed VLP method. Because the linear interpolation (LI) method is simple and the reconstruction accuracy is high [31]. Therefore, we choose the LI method to reconstruct the RSS values of the remaining grid points. When $S = 20$ cm and $R_{SR} = 40\%$. The results of repeated execution 100 times are shown in Fig. 19. Fig. 19 shows that after LI, the CDF of the proposed VLP method is optimal. The average PEs before and after LI are shown in Table IV. Table IV shows that compared with the real dense database, the average PE of the seven VLP methods in the dense database after interpolation are all larger. The main reason is that LI method uses simple linear weighting to reconstruct the RSS value of the interpolated location, so when the training locations are uniformly distributed, LI method can achieve higher RSS reconstruction accuracy. When $R_{SR} = 40\%$,

TABLE IV
THE AVERAGE PE BEFORE AND AFTER LINEAR INTERPOLATION

VLP Method	Sparse database	Dense database after interpolation	Real dense database
WKNN-ED	11.28 cm	8.77 cm	4.35 cm
WKNN-SCD	9.83 cm	7.92 cm	3.78 cm
MLP	6.45 cm	5.73 cm	3.89 cm
LRML	9.01 cm	8.21 cm	7.15 cm
OMP	7.01 cm	5.98 cm	3.84 cm
RWL ₁ M	7.98 cm	6.46 cm	3.93 cm
Proposed	4.31 cm	3.85 cm	1.74 cm

the distribution of training positions is uneven and sparse, which reduces the weight accuracy of the LI method and makes the LI method would not work as well. However, compared with the sparse database before LI, the average PE of the seven VLP methods is reduced. Table IV shows that the average PE of the proposed VLP method is the smallest in three different cases. In the dense database after interpolation, compared with WKNN-ED, WKNN-SCD, MLP, LRML, OMP, and RWL₁M methods, the average PE of the proposed VLP method can be reduced by 56.10%, 51.39%, 32.81%, 53.11%, 35.62%, and 40.40%, respectively.

V. CONCLUSION

IPS based on VLC and location fingerprinting can achieve cm-level positioning accuracy, which has become an important candidate for high precision positioning. However, the fingerprinting weight calculation method is still not accurate enough, resulting in a large PE in the case of low-density and non-uniform location fingerprinting distribution. To solve this problem, an accurate VLP method based on location fingerprinting and meta-heuristic is proposed, and all possible cases of uneven distribution of location fingerprinting are considered by introducing the random selection rate. The effectiveness of the proposed VLP method is compared with six other VLP methods by simulation and experiment. Experimental results show that when the random selection rate is only 50%, the average PE of the proposed VLP method is 3.11 cm, which is at least 41.87% lower than the other six VLP methods. The proposed VLP method has good robustness and can be applied to positioning scenarios with different fingerprinting distributions. Experimental results also show that the method of reducing data collection can also be applied to the proposed VLP method. The research results can provide a new research idea for the fingerprinting weight calculation method.

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